

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 28

HALF ADDER

AIM: To design and implement a Half Adder circuit using Logisim simulator.

ALGORITHM

1. Open the Logisim simulator.
2. Place two Input Pins and name them A and B.
3. Place an XOR gate for obtaining the Sum.
4. Connect inputs A and B to the XOR gate.
5. Place an AND gate for obtaining the Carry.
6. Connect inputs A and B to the AND gate.
7. Connect the XOR output to Sum (S) and AND output to Carry (C).
8. Test the circuit for all possible input combinations.
9. Verify the outputs with the truth table.

Truth Table:

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

RESULT:

